Chirantha K M

Python Engineer | Applied AI/ML | Quantitative Research
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SUMMARY

Python software and applied AI/ML engineer experienced in building production data pipelines, backend APIs, full-stack applications, and statistical ML solutions. My work spans financial-data reconciliation, end-to-end technical delivery, and ongoing research into LLM output validation and hallucination detection.

SKILLS

Programming & Data: Python, SQL, Go, C++, Pandas, NumPy, Data Structures & Algorithms, Object-Oriented Programming
Backend & Platforms: FastAPI, REST APIs, Kafka, Redis, WebSockets, JWT / RBAC, PostgreSQL, MongoDB
Applied AI & Research: Scikit-learn, XGBoost, Statistical Analysis, Anomaly Detection, Feature Engineering, Predictive Modelling, LLM Reliability Research
Infrastructure & Quality: Docker, Kubernetes, GitHub Actions, pytest, Prometheus, Grafana, GCP Cloud Run, AWS (working knowledge), Linux

EXPERIENCE

Quant Analyst & Tech

Reverb Inc. | Remote | May 2026 - Sep 2026
- Translated quantitative research questions into reproducible data workflows, experiments, and Python-based solutions; refined approaches using results and stakeholder feedback.
- Owned data preparation, quantitative analysis, debugging, and technical delivery from problem definition through implementation in collaboration with stakeholders.

Full-Stack Engineer & Technical Lead (Intern)

MSR Solutions | Bengaluru, India | Dec 2025 - May 2026
- Independently shipped production web, mobile, admin, and CRM applications, designing backend APIs, data integrations, and maintainable services.
- Developed an ML-powered estimation feature, translating quantitative cost and timeline models into an end-user product.
- Led architecture, implementation, testing, and deployments through rapid iterations with technical and business stakeholders.

Backend / Data Engineer Intern

SuperBuilt AI | Remote | Feb 2025 - May 2025
- Developed Python and SQL financial-data pipelines processing 50,000+ records daily; implemented reconciliation, validation, idempotent retries, and dead-letter handling to achieve 99.4% reliability.
- Improved PostgreSQL reconciliation queries from 8s to 900ms using composite indexes, partitioning, and schema optimization.
- Implemented statistical anomaly detection with Z-score and Isolation Forest; exposed Prometheus metrics and investigated data-quality failures.

Data Science Intern

YBI Foundation | Bengaluru, India | Jan 2024 - Feb 2024

- Built a reproducible Python ML pipeline for ingestion, feature engineering, training, and evaluation, with 78% pytest coverage.
- Reduced debugging MTTR from 4 hours to under 45 minutes using structured logging and Grafana dashboards.

PROJECTS

Real-Time Statistical Anomaly Detection Engine

- Processed 500 transactions/sec through Kafka, combining Z-score, IQR, and Isolation Forest with PostgreSQL persistence, API endpoints, and latency monitoring.

Stack: Python, FastAPI, Kafka, Scikit-learn, PostgreSQL, Prometheus, Grafana, Docker

Production REST API - Task & Team Management

- Achieved sub-100ms p95 latency under 500 concurrent users and 84% pytest coverage with JWT/RBAC, Redis rate limiting, and automated testing.

Stack: FastAPI, PostgreSQL, Redis, JWT, Docker, pytest, GitHub Actions

Real-Time Event Streaming & Notification Service

- Handled 1,000+ events/min with Kafka consumer groups, dead-letter routing, idempotency, Redis, and consumer-lag monitoring.

Stack: Python, FastAPI, Kafka, WebSockets, PostgreSQL, Redis, Prometheus

Ovia - Predictive Modelling Research

- Developed a reproducible XGBoost workflow for feature processing, evaluation, and FastAPI inference with PostgreSQL; research accepted for IEEE NELEX 2026.

Stack: Python, XGBoost, FastAPI, PostgreSQL, Docker

RESEARCH & PUBLICATIONS

Output Validation & Hallucination Detection for LLM Inference Systems

2025 - present | Ongoing research

- Research on LLM inference output validation, hallucination detection, model failure analysis, and reliability evaluation.

Ovia: XGBoost-Based Predictive Modelling for Women's Wellness Tracking

2026 | IEEE NELEX 2026 | Conference accepted

- Predictive modelling research using XGBoost, feature processing, evaluation, and API inference.

Fast Fashion's Impact on Youth & Sustainability

2024 | Asia Journal of Ethics and Sustainability | Journal publication

EDUCATION

B.Tech - Computer Science & Business Systems

Jain (Deemed-to-be University) | Bengaluru, India | 2022 - 2026

- GPA: 8.3/10
- Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Operations Research, Business Analytics, Computer Networks.

CERTIFICATIONS

2024 | Meta Backend Developer Professional Certificate | Meta / Coursera

2024 | Apache Kafka for Beginners - Developer Certificate | Confluent
2024 | Docker for Developers | Docker Inc.
2024 | GitHub Actions CI/CD | GitHub Skills
2023 | Google Cloud Fundamentals | Google Cloud / Coursera
2023 | Linux Command Line Essentials (NDG) | Cisco NetAcad
2023 | Machine Learning with Python | IBM SkillsBuild

ACHIEVEMENTS & LEADERSHIP

2025 | Top 5 Finalist - Indegene Hackathon: 2,000+ teams; real-time anomaly detection backend.
2024 | Open-Source Contributor: Backtrader & FastAPI-pagination.
2023 | TCS IT Quiz - National Level Representative: Karnataka state selection.
2023 | Founder & Technical Lead - Echo Club: FinTech-focused student community, Jain University; 2023-present.
2022 | Founding Member & Tech/PR Lead - Tech Council SCSE: Led technical and PR operations for three years.
2022 | Technical Lead - Cultural Club: Managed AV infrastructure for large-scale campus events.

CONTACT

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